

Student Mental Health & Social Media Analytics

Business Intelligence Report

Executive Summary

The objective of this project was to analyze the relationship between social media usage and student well-being, with a particular focus on sleep patterns, stress levels, and behavioral indicators. Using SQL, Power BI, and Python, the dataset was explored to identify trends, correlations, and actionable insights that support data-driven decision-making in educational institutions.

The analysis revealed a noticeable relationship between increased social media usage, reduced sleep duration, and higher stress levels among students. These findings highlight the importance of digital wellness initiatives and proactive student support programs.

Business Problem

Educational institutions are increasingly concerned about the impact of social media on student mental health and academic performance. While social media offers opportunities for communication and learning, excessive usage may contribute to poor sleep quality, increased stress, and reduced overall well-being.

This project aims to answer the following questions:

- How much time do students spend on social media?
- Does social media usage affect sleep duration?
- Is there a relationship between social media usage and stress levels?
- Which student groups appear to be at higher risk?
- What actions can educational institutions take based on the findings?

Objectives

- Analyze student social media usage patterns.
- Measure the relationship between sleep duration and social media usage.
- Evaluate stress levels across different student groups.
- Identify high-risk behavioral patterns.
- Develop interactive dashboards to support decision-making.

Tools & Technologies

- SQL Server
- Power BI
- Python (Pandas)

Data Preparation

The dataset was cleaned and validated before analysis.

The following activities were performed:

- Missing value assessment
- Data type validation
- Duplicate record checks
- Feature standardization
- Categorization of social media usage levels

The cleaned dataset was loaded into SQL Server for analytical querying and reporting.

Key Performance Indicators (KPIs)

The following KPIs were developed:

- Total Students
- Average Daily Social Media Usage
- Average Sleep Duration
- Average Stress Level
- High-Risk Student Percentage
- Platform Usage Distribution

These KPIs provide a high-level overview of student wellness and engagement trends.

Analysis and Findings

1. Social Media Usage Analysis

The analysis showed varying levels of social media engagement among students. A significant portion of students reported multiple hours of daily social media activity.

Key questions investigated:

- Which platforms are used most frequently?
- How does daily usage vary among students?

2. Sleep Duration Analysis

Sleep duration was analyzed across different usage groups.

Findings:

- Students with higher social media usage generally reported fewer hours of sleep.
- Late-night usage patterns were associated with reduced sleep duration.

Business Insight:

Excessive screen time may negatively affect sleep quality and recovery.

3. Stress Level Analysis

Stress levels were evaluated across student segments.

Findings:

- Students with higher social media usage reported higher levels of stress.
- Reduced sleep duration was often associated with elevated stress indicators.

Business Insight:

Sleep and social media behavior may act as early indicators of student well-being concerns.

4. Risk Segmentation

Students were categorized into risk groups based on:

- Social media usage
- Sleep duration
- Stress levels

Risk Categories:

- Low Risk
- Moderate Risk
- High Risk

This segmentation enables institutions to focus support efforts on the most vulnerable student populations.

Dashboard Overview

The Power BI dashboard includes:

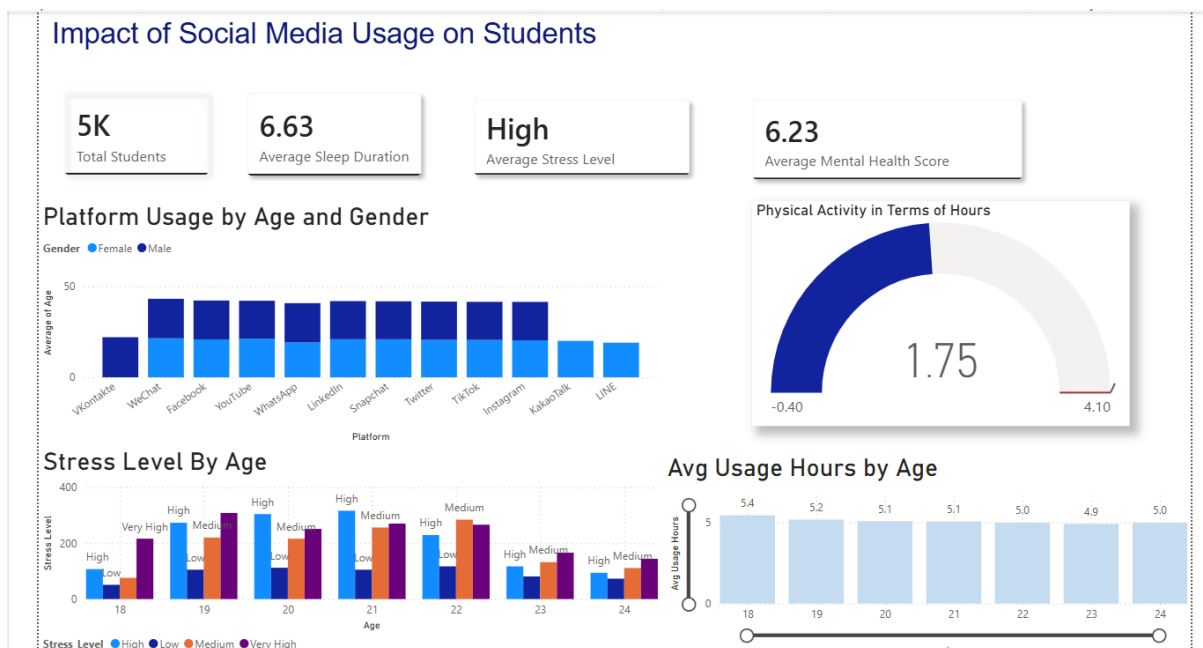


Figure 1: Social Media Impact

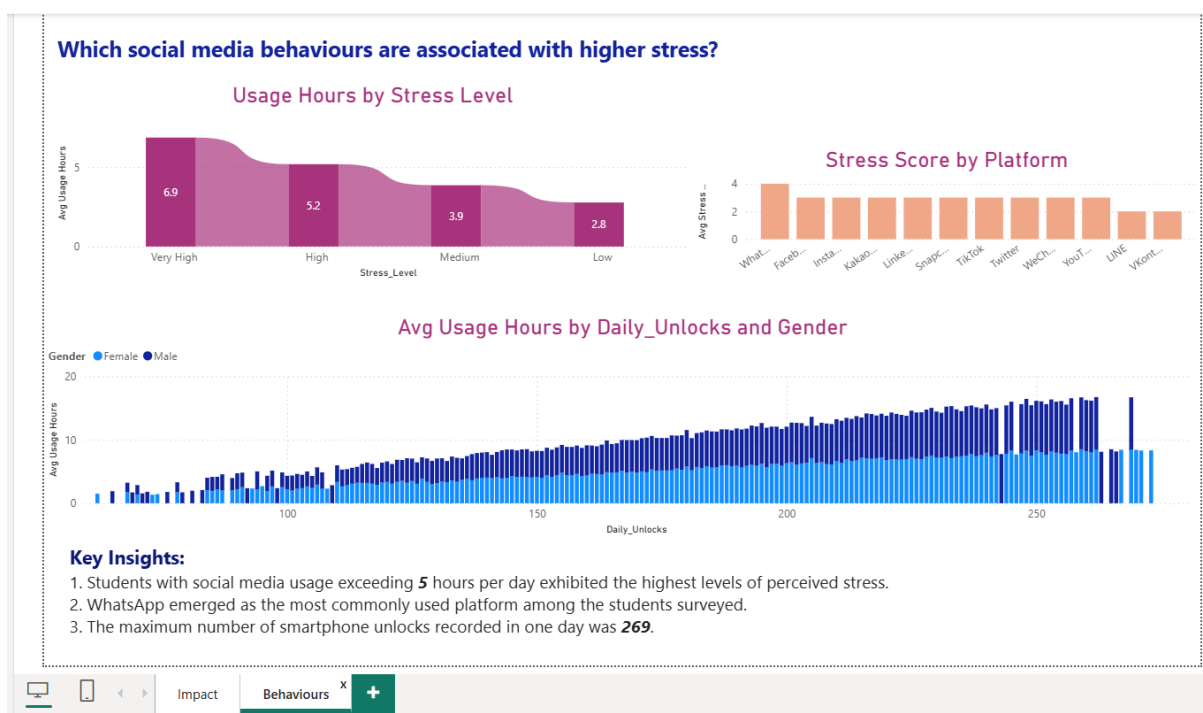


Figure 2: Social Media Behavior

Executive Dashboard

- Overall KPIs
- Student population overview

Social Media Analytics

- Platform usage trends by age and gender
- Daily usage patterns and Unlocks

Sleep & Wellness Dashboard

- Average Physical Activity among Students
- Stress Levels by Age

Stress Analysis Dashboard

- Stress level by usage hours
- Stress Score by Platform

Recommendations

Based on the analysis, the following actions are recommended:

1. Promote Digital Wellness Programs

- Educate students on healthy social media habits.

2. Increase Mental Health Awareness

- Conduct workshops and awareness campaigns.

3. Monitor High-Risk Groups

- Use behavioral indicators to identify students requiring support.

4. Encourage Healthy Sleep Habits

- Promote sleep hygiene and reduced nighttime screen exposure.

5. Leverage Data-Driven Student Support

- Continuously monitor wellness KPIs through dashboards and reporting.

Conclusion

This project demonstrates how Business Intelligence tools can be used to transform raw student data into meaningful insights. By combining SQL, Power BI, and Python, the analysis provides a comprehensive view of the relationship between social media usage and student well-being.

The findings support data-driven decision-making and highlight opportunities for educational institutions to improve student wellness through targeted interventions and ongoing monitoring.